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Analysis of the seismic behavior of CFRP-strengthened seismic-damaged composite steel-concrete frame joints

Chengxiang Xu [a], Sheng Peng [a,*], Xiaoqiang Liu [b], Chenfei Wang [a], Qiqi Xu [c,**]

[a] Department of Civil Engineering, School of Urban Construction, Wuhan University of Science and Technology, Wuhan, 430065, PR China
[b] School of Architectural Engineering, Weifang University of Science and Technology, Weifang, 262700, PR China
[c] Wuhan Building Material Industry Design and Research Institute Co. Ltd, Wuhan, 430200, PR China

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ABSTRACT

An experimental investigation was conducted to examine the seismic behavior of three seismic-damaged composite steel-concrete frame joints strengthened with carbon fiber-reinforced polymer (CFRP) sheets, as well as of one original composite steel-concrete frame joint. Effects of the degree of seismic damage and the CFRP reinforcement on shear capacity, ductility, energy dissipation, and stiffness degradation were critically researched. Based on an existing code, an expression to predict the shear strength of seismic-damaged composite steel-concrete frame joints strengthened with CFRP is presented; the influence of the degree of seismic damage and amount of CFRP on the shear strength is considered in this expression. Results of the proposed expression and the test results were compared. The results revealed that the ultimate bearing capacity, ultimate displacement, and ductility of the specimens improved after CFRP reinforcement. There was good agreement between predicted results from the shear strength model and the test results with regard to the degree of seismic damage and amount of CFRP.

1. Introduction

Composite steel-concrete frame structures are some of the most widely used seismic structure forms at present [1]. During seismic events, the building structure is damaged in varying degrees and the repairs and restoration are performed depending on the degree of earthquake damage [2,3]. However, not all materials used in repairs and restoration are earthquake resistant. The choice of material directly affects the earthquake resistance of the building and structural collapse and shear failure may result [4]. This area represents one of the most important research topics focused on the prevention of shear failure of seismic-damaged composite steel-concrete structures.

Carbon fiber-reinforced polymer (CFRP) sheets have many excellent properties, such as high-strength, light-weight, corrosion-resistance, diamagnetic resistance, and ease of fabrication [5–7]. Research on CFRP sheets and their engineering applications have been conducted in China and abroad and satisfactory research progress has been made. Yurdakul and Avsar [8] studied the current conditions and future prospects of restoration and repair of seismic-damaged concrete frames and pointed out that the repair and restoration of damaged structures should achieve both “damage repair” and “seismic performance improvement”. Based on existing codes, Chang et al. [9] designed and manufactured concrete frame joints and simulated seismic damage by pre-damage loading. The seismic behavior of the damaged concrete frame joints strengthened with CFRP was investigated. Tsonos [10] constructed a frame joint specimen, conducted a seismic damage simulation, and studied the influence of different restoration methods on the seismic behavior of the seismic-damaged joints. Xu et al. [11] designed and manufactured four one-half scale models of seismic-damaged exterior joints in a composite consisting of concrete-filled square steel tubular (CFSST) columns strengthened with CFRP and steel beams; the authors studied the effectiveness of the seismic-damaged joints strengthened with CFRP and the reinforcement for different levels of seismic damage.

Common failure modes of frame joints include the bending failure of the beam ends, bending failure of the ends of columns, and shear failure of the core area of joints among others [12,13]. The theoretical analysis of the shear failure in the core area of joints is becoming increasingly important [14]. Mazza [15] and Zamani and Shariatmadar [16] studied the shear properties and damage characteristics of reinforced concrete

* Corresponding author.
** Corresponding author.
E-mail addresses: pe_sh@sina.com, pengsheng@wust.edu.cn (S. Peng), 491829546@qq.com (Q. Xu).

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(RC) beam-column joints strengthened by CFRP. Elsouri and Harajli [17] tested four frame joints of concrete reinforced with CFRP under reversed cyclic loading and compared the measured bearing capacity of the joints with calculated bearing values based on codes to verify the effectiveness. Karayannis and Sirkelis [18] studied the seismic behavior of four concrete beam-column joints strengthened with short glass fiber-reinforced (SGFR) polymers and hybrid fiber-reinforced polymers (HFRP) under reversed cyclic loading. A formula for calculating the shear bearing capacity of the joints strengthened with fiber-reinforced polymers (FRP) was proposed and the impact mechanism of the shear strength of the joints was determined.

At present, the restoration of damaged frame joints with CFRP mainly includes the use of RC beam-RC columns and CFSST column-RC beams; however, only few research studies have been published on the seismic behavior and shear bearing capacity of seismic-damaged composite steel-concrete frame joints strengthened with CFRP. Hence, in this study, we investigate the behavior of seismic-damaged composite steel-concrete frame joint strengthened with CFRP under reversed cyclic loading. Based on the existing code, we propose an expression of the shear strength of seismic-damaged composite steel-concrete frame joints strengthened with CFRP that considers the effects of the degree of seismic damage and amount of CFRP on the shear strength. The results of the proposed expression and the test results are compared.

2. Experimental methods

2.1. Test specimens

The middle joints of the bottom plane of the plane frame were chosen as the areas of interest. Four composite steel-concrete frame joints were designed and built at a 1/2 scale and were tested under the combined action of an axial load and reverse circulation lateral displacement. The distance between the counter bending points of the beam was 2000 mm and the distance between the counter bending points of the column was 1570 mm. The cross-sectional area of the beam was 200 mm × 320 mm and the column section area was 250 mm × 250 mm. The material of the column cross-section is I14 steel (model Q235B) and the steel ratio of the specimen was 3.44%. The longitudinal reinforcement of the specimen was HRB400 and the cross-section reinforcement ratio of the specimen was 0.98%. The core area of the specimen was the column-through type. The connecting plate and stiffener were welded with section steel and had a thickness of 5 mm. The concrete cover was 20 mm and the effective height of the frame joint was $H_0 = 1570$ mm, as shown in Fig. 1. The average compressive strength of the measured concrete cube was 39.6 Mpa; the concrete was poured from the same batch and was cured for 28 days.

The data shown in Table 1 indicate that the specimens SRCC-0, SRCC-1, SRCC-2, and SRCC-3 were under a constant axial load ratio and axial-load and had the same concrete strength. Specimen SRCC-0 was not strengthened and damaged and served as the control sample. Specimen SRCC-1 was strengthened with the CFRP sheets and was undamaged. An interlayer displacement angle of 1/100 was used to simulate moderate damage for the specimen SRCC-2. An interlayer displacement angle of 1/50 was used to simulate severe damage for the specimen SRCC-3. Specimens SRCC-2 and SRCC-3 were strengthened with the CFRP sheets. The details of the specimens are shown in Fig. 1.

2.2. Material properties

The mechanical properties of the materials, including the concrete, longitudinal reinforcement, stirrup, section steel, CFRP, and mucilage are listed in Table 2.

2.3. Reinforcement of specimens

The rehabilitation of the specimens was carried out by using the GB50367-2013 guidelines. An HP-12K-300 press was used to strengthen the specimens. The concrete rehabilitation methods were as follows [19]: (1) two layers of L-shaped CFRP were attached at 450 mm from the end to the upper and lower side of the beam end and extended 150 mm from the end of the column. The CFRP was 180 mm wide and 600 mm long. (2) The core area of the joint was pasted with one layer of CFRP with a length of 520 mm and a width of 320 mm and extended 100 mm from the end of the beam; one layer of CFRP with a length of 520 mm and a width of 250 mm was attached vertically and extended 100 mm from the end of the column bottom. (3) Two layers of CFRP were attached to the ring hoop at 300 mm at the column and near the joint. (4) At 450 mm from the beam end near the joint, the CFRP ring hoop with a width of 100 mm was anchored. Fig. 2 shows the details and the strain gauge layout; the strain of the longitudinal reinforcement (S1–S2) was measured.

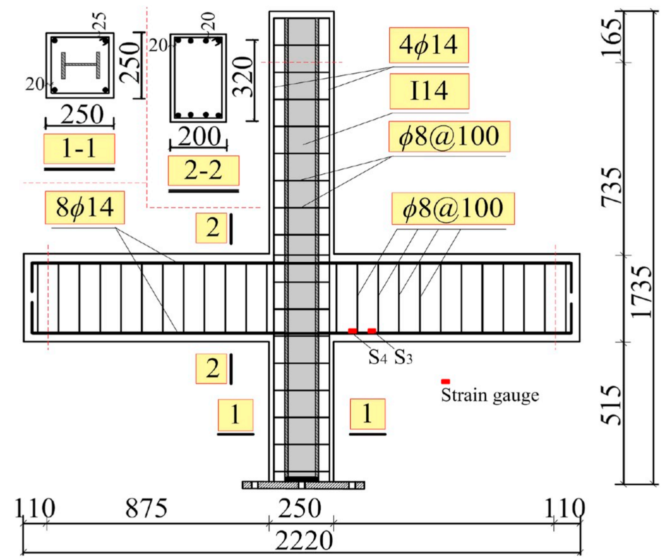


Fig. 1. Details of specimen (Note: dimensions are in mm).

Table 1
Parameters of specimens.

Specimen	Damage degree	n	Axial-load (kN)	Concrete strength grade	Reinforcement method
SRCC-0	–	0.40	750	C40	–
SRCC-1	–	0.40	750	C40	CFRP reinforcement
SRCC-2	Moderately damaged	0.40	750	C40	CFRP reinforcement
SRCC-3	Severely damaged	0.40	750	C40	CFRP reinforcement

2.4. Test device and loading system

The joints were built and tested at the Civil Engineering Experiment Center of Wuhan University of Science and Technology. All specimens were checked after 28 days. A hinge was attached at the bottom end of the column and the two ends of the beam articulated by means of a connecting rod, thereby creating a movable horizontal support. The vertical axis load at the upper end of the column was implemented by a hydraulic jack and the load remained constant. The reversed lateral cyclic loading was applied by a servo-controlled hydraulic actuator at the upper column end [20]. The range of the displacement control was ±150 mm. During the horizontal loading process, the hydraulic jack moved along with the sliding trolley.

The vertical load was applied by the hydraulic jack first and

Table 2
Materials properties.

Material	Specification	Tensile strength f_t/MPa	Compressive strength f_c/MPa	Yield strength f_y/MPa	Ultimate strength f_u/MPa	Elastic modulus E/MPa
concrete	150 mm length cube	–	39.6	–	–	–
I14 steel	Q235B	–	–	312.4	434.2	2.01×10^5
longitudinal bar	HRB400	–	–	405.6	536.8	2.17×10^5
hoops	HPB300	–	–	312.3	443.1	2.10×10^5
CFRP	CJ300-I	3560	–	–	–	2.50×10^5
Mucilage	CFRP mucilage	42	–	–	–	2476
	WSJ mucilage	30	–	–	–	1563

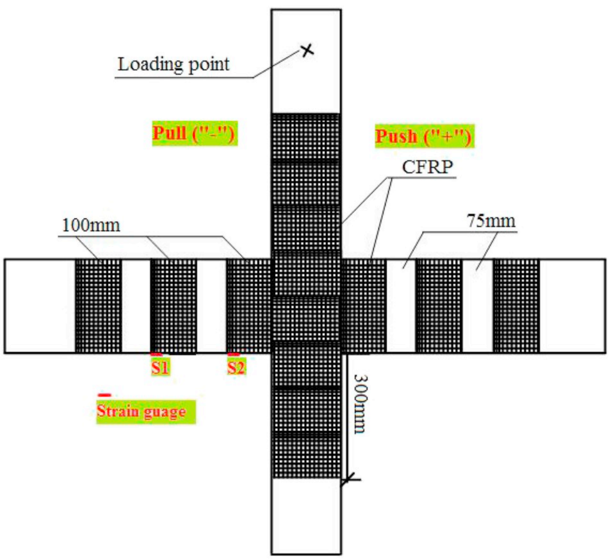


Fig. 2. Schematic diagram of strengthening of CFRP sheets.

remained constant at a preset value; second, the horizontal cyclic loading was applied by the horizontal actuator. A twin control method of load and displacement was applied to the horizontal load and the yield of the specimen was marked by the yield of the beam's longitudinal reinforcement. The load control occurred before the specimens yielded and the displacement control occurred after the specimens yielded. The specimens were submitted to one cycle before yielding and three successive cycles after yielding. When the bearing capacity of the specimen was less than 85% of the ultimate load or the specimens were not suitable for continued loading, the test was stopped.

The pre-damage of the specimen was controlled by the displacement and the reversed lateral cyclic loading was carried out in steps of 3 mm. As shown in Fig. 3 (b), the specimen was re-tested one time at each stage before yielding [21]. After yielding, the specimen was re-tested three times at each stage until the pre-damage displacement was reached [22]. Specifically, after the horizontal position of the predetermined distance was reached, the oil pump was discharged slowly until the horizontal position was reached; then it was pulled back to the predetermined distance, and the oil pump was discharged slowly until the horizontal position was reached, and so on. Subsequently, the cracks were repaired with a water-extended polyester (WEP) injection sealant. The crack width was measured using a concrete crack width measuring instrument. The pre-damage displacement corresponding to moderate and severe earthquake damage was 20 mm and 45 mm, respectively. The columns were laterally subjected to a predetermined cyclic displacement history as indicated in Fig. 3(a). The test setup and loading system are shown in Fig. 3(b).

2.5. Strain gauge layout

The test measured the crack width of the concrete, the horizontal displacement and load at the upper end of the column, the strain of the longitudinal reinforcement and stirrup of the column and beam, the strain of the section steel flange and web of the joint core, and the strain of the concrete surface and CFRP surface. A BZ2205C static resistance strain instrument was used for the measurements [22]; the arrangement of the strain gauge is shown in Figs. 1 and 2.

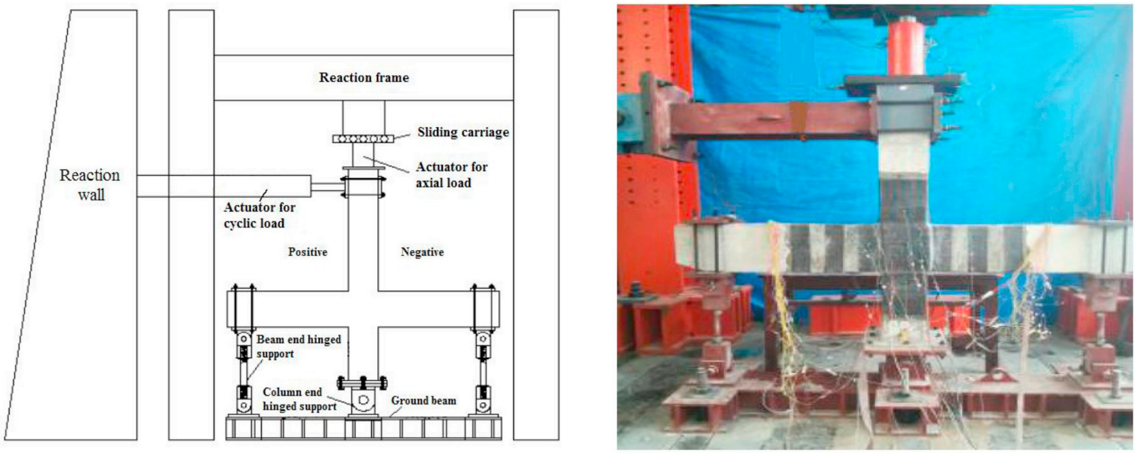
3. Experimental results

3.1. Experimental phenomena and failure modes

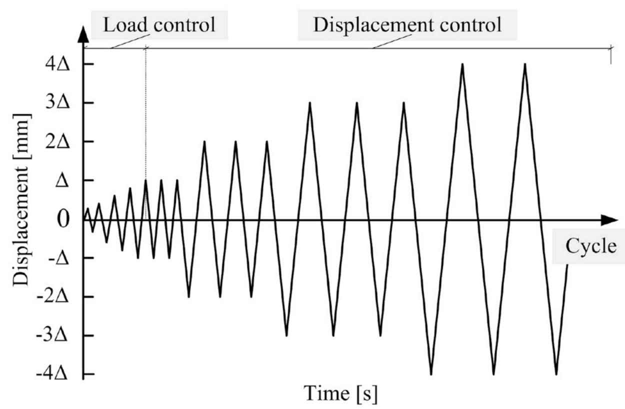
A vertical load of 750 kN was applied first using the hydraulic jack and remained constant; the horizontal load was applied after the instruments were working normally. To facilitate the description, during the experiment, the actuator operates as “pull equals positive values and push equals negative values”.

The specimen SRCC-0 was the control specimen and was loaded to induce damage but it was not reinforced. The fracture distribution of the specimen SRCC-0 is shown in Fig. 4 (a). During the±10 kN cyclic loading process, the cyclic loading was increased to +8.5 kN, which resulted in the appearance of the first cracks with a width of 0.1 mm. During the±30 kN cyclic loading process, when the cyclic loading was increased to +22.3 kN, the second and third sets of cracks appeared. As the displacement decreased, the cracks closed completely. During the ±50 kN cyclic loading process, when the cyclic loading increased to +42 kN, the fourth set of cracks appeared. Meanwhile, the first, second, and third sets of cracks gradually extended. When the horizontal displacement increased to −18 mm, the fifth set of cracks appeared. The strain of the longitudinal reinforcement of the beam was greater than the yield strain and the specimen yielded. During the first cyclic loading when the horizontal displacement increased to ±30 mm, the sixth set of cracks appeared; the original cracks continued to develop and the cracks extended throughout the joints of the specimens. During the first cyclic loading with a horizontal displacement of ±45 mm, as the horizontal displacement increased to −42 mm, the seventh set of cracks appeared and the concrete began to peel in the lower right beam. In the first cyclic loading process with a horizontal displacement of ±60 mm, concrete spalling occurred on the upper part of the right beam that was in contact with the column. The concrete on the lower part of the right beam was crushed, the beam hoop reinforcement was exposed, and the crack width of the beam increased to 4 mm. Subsequently, the bearing capacity of the specimen dropped to 85% of the ultimate bearing capacity and the test stopped. The specimen failure modes are shown in Fig. 4(a).

The specimen SRCC-1 was not pre-damaged but was strengthened by the CFRP and was directly loaded to induce damage. The surface of the concrete was wrapped with CFRP; therefore, the distribution of the cracks could not be observed easily. The specimen yielded when the horizontal displacement increased to −18 mm. During the first cyclic loading process of ±30 mm, when the horizontal displacement increased to −25 mm, the colloid fragile site of the CFRP emitted a breaking sound. During the second cyclic loading process of ±45 mm, when the



(a) Test set-up for cyclic loading test



(b) Loading system for horizontal displacement

Fig. 3. Test set-up and loading system.

(a) SRCC-0 (b) SRCC-1 (c) SRCC-2 (d) SRCC-3

Fig. 4. Destructive phenomena of specimens.

horizontal displacement increased to +41 mm, cracks appeared on the surface of the concrete on the side of the left beam, which was not wrapped with the CFRP sheets. The sound of the breaking of the colloids was loud and the CFRP of the lower surface of the right beam tore slightly. During the first cyclic loading process of ±60 mm, the CFRP exhibited serious tearing and the bearing capacity began to decline. During the first cyclic loading process of ±75 mm, the CFRP on the lower surface of the right beam broke. Subsequently, the bearing capacity of the specimen dropped to 85% of the ultimate bearing capacity and the test stopped. The failure modes of the limit state at the bottom end of the columns of the seismic-damaged CFRP-strengthened specimens, including SRCC-1, SRCC-2, and SRCC-3, are shown in Fig. 4(b)–(c).

The specimen SRCC-2 was the moderately damaged specimen that was strengthened with CFRP and was directly loaded to induce damage. When the cyclic loading increased to ±60 kN, a breaking sound was heard and originated from the CFRP. When the horizontal displacement increased to −18 mm, the specimen yielded. During the first cyclic loading of ±30 mm, the crunchy sound of the CFRP became more obvious. When the horizontal displacement increased to −25 mm, cracks appeared on the surface of the concrete on the side of the left beam, which was not wrapped with the CFRP sheets. During the first cyclic loading of ±45 mm, when the horizontal displacement increased to +42 mm, the CFRP on the upper surface of the right beam was slightly damaged and the surface of the concrete separated; the CFRP on the lower surface was partially torn and the sound of the formation of brittle cracks was heard. When the horizontal displacement increased to ±78 mm, the bearing capacity of the specimen dropped to 85% of the ultimate bearing capacity and the test stopped.

The specimen SRCC-3 was the severely damaged specimen strengthened with the CFRP and was directly loaded to induce damage. When the cyclic loading increased to +60 kN, a breaking sound was heard and originated from the CFRP. When the horizontal displacement increased to −18 mm, the specimen yielded. During the first cyclic loading of ±30 mm, when the horizontal displacement increased to +28 mm, the CFRP on the lower surface of the right beam was slightly damaged and the sound of the formation of brittle cracks became louder. During the first cyclic loading of ±60 mm, when the horizontal displacement increased to +51 mm, the bearing capacity began to decline. During the first cyclic loading of ±75 mm, the CFRP on the lower surface of the right beam broke. Subsequently, the bearing capacity of the specimen dropped to 85% of the ultimate bearing capacity and the test stopped.

3.2. Hysteresis loops

The hysteresis envelope curves of the horizontal load displacement are shown in Fig. 5. The conclusions are as follows:

After yielding, the residual deformation of the specimens increased continuously and the area of the hysteresis loop increased. During loading, the slope of the hysteresis envelope curves decreased with the increase in the displacement. During the unloading process, the displacement lag was obvious and the curve was steep.

The fluctuations of the hysteresis curves of the strengthened specimens has decreased. The ultimate displacement increased significantly and the hysteresis loop area was larger. This indicates that the deformability and energy dissipation were clearly improved by the CFRP reinforcement.

Compared with specimens SRCC-1 to SRCC-3, the hysteresis curve gradually decreased with the increase in the degree of seismic damage. The degree of damage influences the effectiveness of the CFRP reinforcement. The smaller the degree of damage, the better the reinforcement effect is.

3.3. Skeleton curve

The skeleton curves of the specimens are shown in Fig. 6. The effect of the CFRP was observed after the specimen had yielded. There was no apparent turning point in the curve, which indicated that the yield of the specimens was a gradual process beginning in a localized area of the specimen and then spreading to the entire specimen. The ultimate bearing capacity was higher for the specimen strengthened with the CFRP than for the control specimen. This shows that the CFRP improved the ultimate bearing capacity of the specimen and the degree of improvement was related to the degree of damage; a higher ultimate

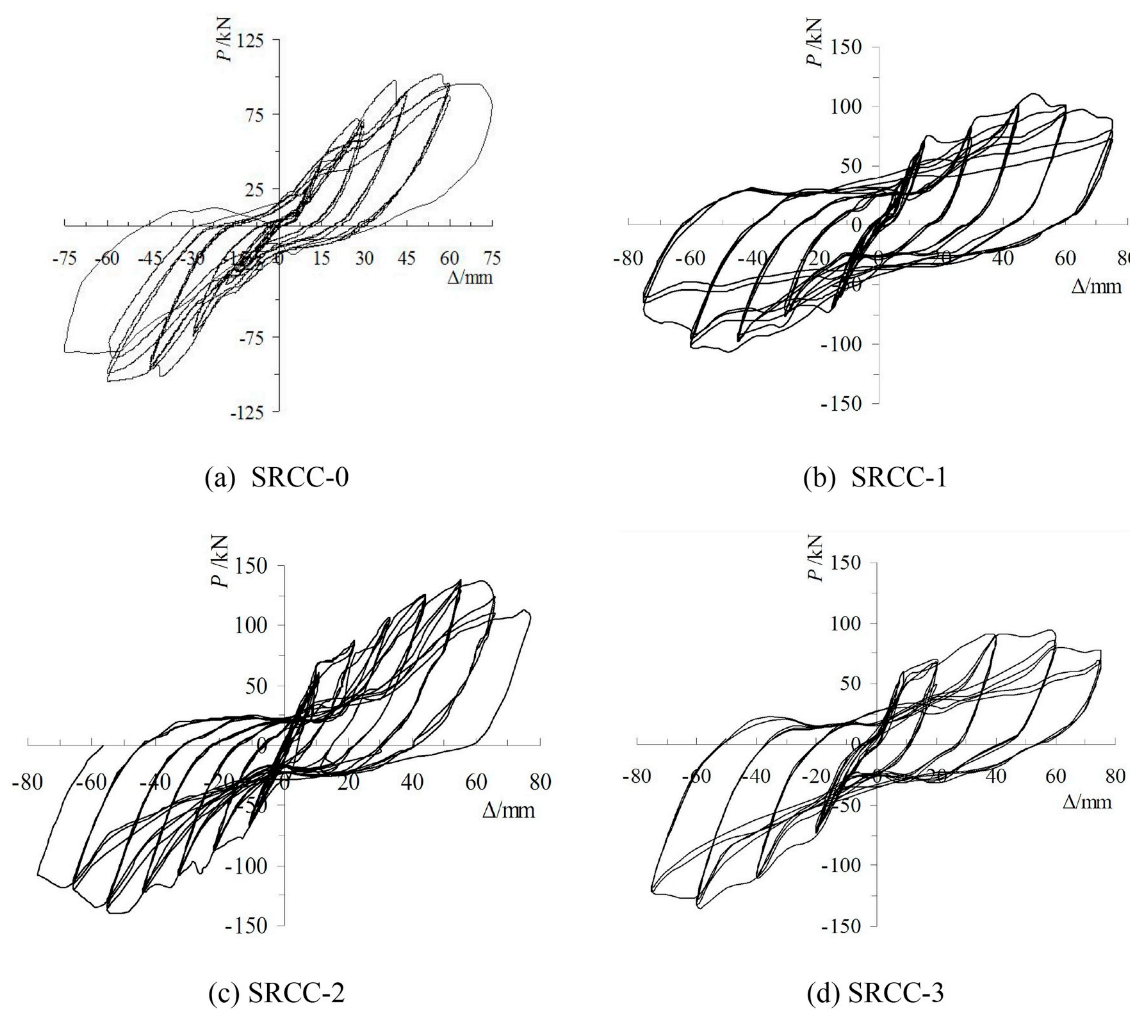


Fig. 5. Hysteretic curves of specimen.

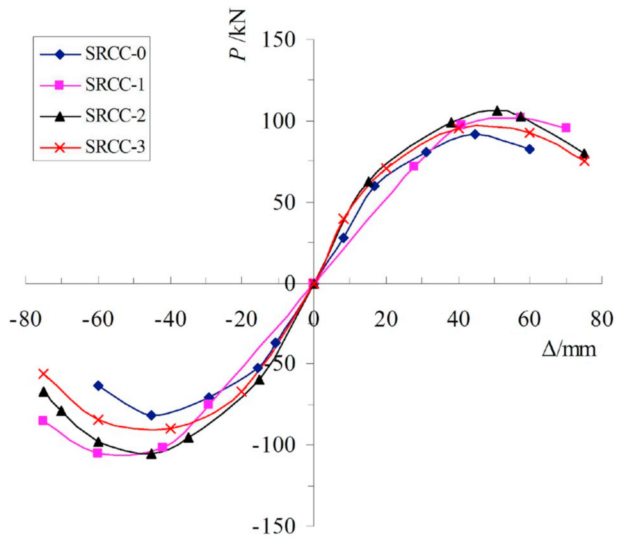


Fig. 6. Skeleton curves of specimens.

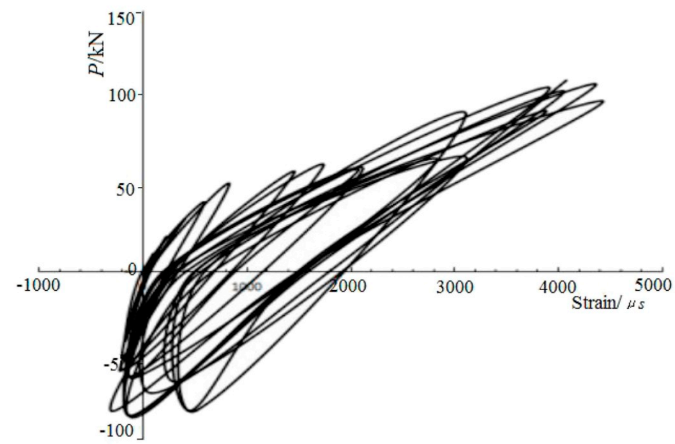


Fig. 8. Hysteretic loops of CFRP.

bearing capacity was observed for the specimens with less initial seismic damage. The ultimate displacement was greater for the specimen strengthened with the CFRP than for the control specimen, which showed that the CFRP effectively improved the ductility of the specimen.

3.4. Strain analysis

Fig. 7 shows the strain-time curves of the SRCC-2 beam longitudinal reinforcement (named S4) and the CFRP of the beam end (named S1). Fig. 8 indicates that the strain of the CFRP was not high during the initial loading stage; the average strain was within 1000 μm and the average strain of the steel reinforcement was 3000 μm. This indicates a stress lag in the CFRP. The CFRP acted as a passive constraint that prevented the deformation of the beam end after the beam's longitudinal reinforcement yielded. Under continuous loading, the steel reinforcement and CFRP began to produce residual strain and the residual strain of the steel reinforcement was larger than the residual strain of the CFRP. During the later loading stage, the CFRP still maintained a high stress level, which showed that the CFRP and concrete were well bonded to the concrete and operated in conjunction with the beam.

The load-strain curve of the CFRP at the beam end of the specimen SRCC-2 (named S2) is shown in Fig. 8. It can be seen that the hysteresis curve is close to a straight line at the beginning of loading; the change in the hysteresis loop was not obvious, the residual deformation was very small, and the CFRP did not play a role at this time. Under continuous loading, the residual deformation of the CFRP increased, the strain increased with the increasing load and the area of the hysteresis loop increased. The results indicated that the CFRP deformed at the beam end, which showed good energy dissipation performance.

3.5. Ductility

The yield point is defined by the "energy equivalent method" and corresponds to the yield load, P_y, the yield displacement, Δ_y; the failure displacement, Δ_u, of the specimen is the displacement limit [5]. P_u is the failure load, Δ_{max} is the maximum displacement, and P_{max} is the maximum load. The displacement ductility factor, μ, is usually defined as the ratio of the ultimate displacement, Δ_u, to the yield displacement, Δ_y. The ultimate displacement, Δ_u, is used as the displacement when the remaining capacity drops to 85% of the maximum applied load. The measured results of the specimens are shown in Table 3.

After the specimen SRCC-1 was strengthened by the CFRP, the average ductility coefficient increased from 3.19 to 4.00, an increase of 25.39%, which indicated that the CFRP effectively improved the ductility of the specimen. Compared with specimen SRCC-1, the average ductility coefficient of the specimens SRCC-2 and SRCC-3 decreased from 4 to 3.85 and 3.36, respectively, a decrease of 3.75% and 16%, respectively; this indicated that the average ductility coefficient of the

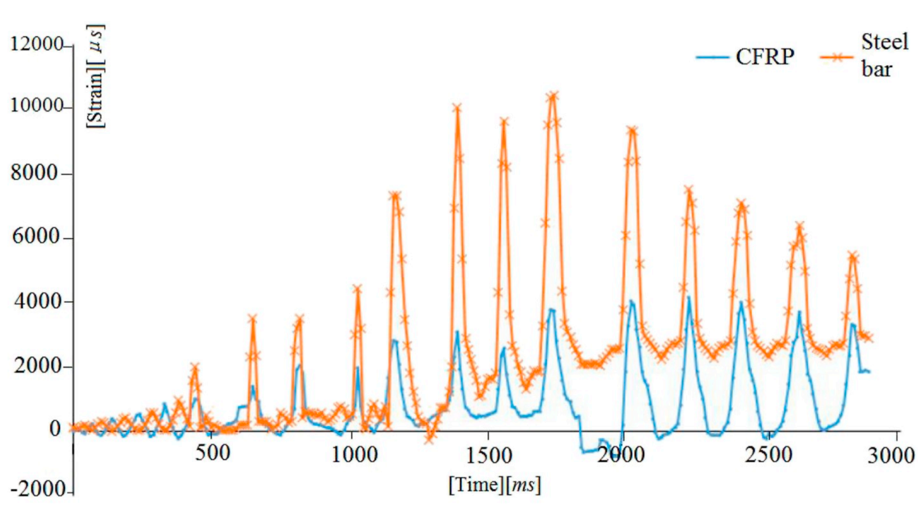


Fig. 7. Strain-time curves.

Table 3
Characteristic points of skeleton curves.

Specimen	Load direction	P_y/kN	Δ_y/mm	P_m/kN	Δ_m/mm	P_u/kN	Δ_u/mm	$\mu = \Delta_u/\Delta_y$
SRCC-0	Positive	63.22	18.12	93.31	44.52	79.31	58.24	3.21
	Negative	67.58	18.87	92.12	44.72	78.30	59.75	3.17
SRCC-1	Positive	60.87	18.81	108.08	50.04	91.87	76.23	4.05
	Negative	74.89	19.78	106.22	51.24	90.29	78.12	3.95
SRCC-2	Positive	67.56	18.55	105.18	50.08	89.40	72.48	3.91
	Negative	76.66	19.48	104.85	51.29	89.12	73.89	3.79
SRCC-3	Positive	70.54	18.36	101.35	51.34	86.15	63.15	3.44
	Negative	71.22	18.84	103.27	55.42	87.78	61.86	3.28

specimen decreased as the degree of damage increased. Compared with specimens SRCC-3 and SRCC-0, the average ductility coefficient increased from 3.19 to 3.36, an increase of 5.33%; this demonstrates the good performance of the CFRP.

3.6. Ultimate bearing capacity

Compared with specimen SRCC-0, the average ultimate bearing capacity of specimens SRCC-1 to SRCC-3 increased by 15.58%, 13.27%, and 10.35%, respectively (Table 3); this indicated that the CFRP improved the bearing capacity of the specimens. The change in the ultimate bearing capacity was not significant for specimens SRCC-1 to SRCC-3, which indicates that the degree of seismic damage had little influence on the bearing capacity of the specimen.

3.7. Stiffness degradation

The stiffness degradation is described by the secant stiffness of the specimens under different loading displacements [23]. The secant stiffness was measured at the peak load in the same loading. Based on the experimental results, the mean value of the secant stiffness for the ith cycle was evaluated using the following ratio:

$$K_i = \frac{\left|P^+_{m,i}\right| + \left|P^-_{m,i}\right|}{\left|\Delta^+_{m,i}\right| + \left|\Delta^-_{m,i}\right|} \tag{1}$$

where K_i is the secant stiffness; the stiffness of each cycle can be normalized with respect to the stiffness of the first cycle. In theory, the elastic stage never undergoes stiffness degradation. The stiffness degradation curve is presented in Fig. 9.

As shown in Fig. 9, as the horizontal displacement increased, the secant rigidity of the specimen gradually decreased. The slope of the stiffness degradation curve became steeper and then gradually

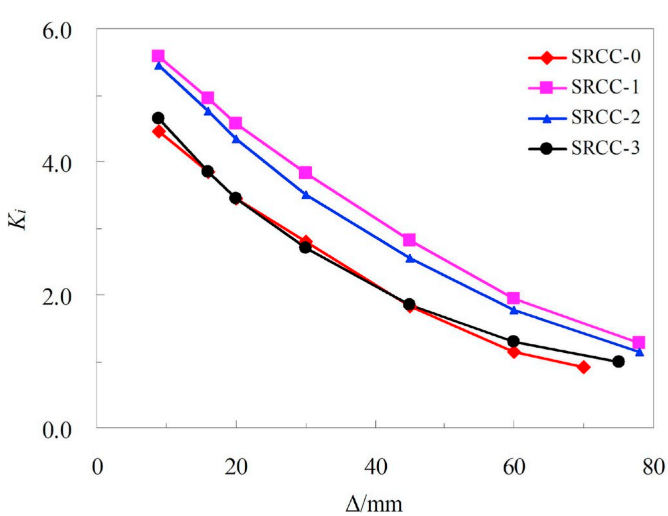


Fig. 9. Viscous damping coefficient of specimens.

decreased. The degree of seismic damage was higher for the CFRP-reinforced specimens and the stiffness degradation occurred more rapidly, which indicated that the degree of seismic damage affected the stiffness degradation of the specimen. The stiffness degradation was lower for the CFRP-reinforced specimens than the control specimen, which indicated that the CFRP sheets effectively prevented damage to the specimen and improved its ability to resist seismic deformation.

4. Analysis of the shear strength of the joint

4.1. Shear strength of the strengthened joint

The shear strength of the seismic-damaged reinforced composite steel-concrete frame joint, V_u, is composed of four parts: the steel webs shear, V_w, the steel flange shear, V_f, the core area of RC frame joint shear, V_{rc}, and the shear force provided by the CFRP, V_{CFRP}.

$$V_u = V_w + V_f + V_{rc} + V_{CFRP} \tag{2}$$

According to the shear capacity analysis of each shear unit, the horizontal shear strength of the two parts of the steel web shear, V_w, and the steel flange shear, V_f, are obtained for the limit state:

$$V_w = \frac{0.58 f^{'}_a t_w h_w}{\lambda - 0.2} \tag{3}$$

$$V_f = 0.05 V_w \tag{4}$$

where f'_a = strength of the component steel after an earthquake. t_w and h_w are the height and thickness of the steel web, respectively. λ is the shear span-to-depth ratio.

The specifications for the design requirements of the shear strength of the frame joints are as follows [19]:

The first case is an aseismic-grade frame of nine degrees of fortification intensity:

$$V_{rc} \le \frac{1}{\gamma_{RE}}\left(0.9\eta_j f_t b_j h_j + f^{'}_{yv} A_{svj} \frac{a - a^{'}_s}{s}\right) \tag{5a}$$

Other cases:

$$V_{rc} \le \frac{1}{\gamma_{RE}}\left(1.1\eta_j f_t b_j h_j + 0.05\eta_j N \frac{b_j}{b_c} + f^{'}_{yv} A_{svj} \frac{a - a^{'}_s}{s}\right) \tag{5b}$$

where γ_{RE} = aseismic adjustment coefficient of the bearing capacity; in this case, γ_{RE} = 0.85 [18]. The height of the cross-section of the core dimension of the frame joint, h_j, is equal to the height of the column section, h_c, in the checking direction. b_j = effective cross-section when checking the width of the core dimension of the frame joint. a = the cross-section of the frame beam has an effective height. When the heights of the cross-section of the two sides of the joint are unequal, the average value is used. f_{yv} = yield strength of the stirrup after an earthquake [1]. A_{svj} = the cross-section dimensions of the hoops. The influence of the coefficient of the orthogonal beam on the joint, η_j, can be obtained as follows: when the floor slab for cast-in-situ, reclosing of middle line of beam and column, the width of the cross-section of the four sides is no less than the width of the cross-section of the side column

1/2 and the height of the orthotropic beam is no less than 3/4 of the high frame beam height; in this case, $\eta_j = 1.5$. For a seismic fortification intensity of nine degrees, we use $\eta_j = 1.25$. The additional parameters are: $\eta_j = 1.0$ [17]. N = design value of the axial load, kN; f_t = concrete tensile strength.

4.2. Shear strength of the CFRP

The shear strength, V_{CFRP}, of the CFRP reinforcement joint core area is related to the type of joint. For the middle node of the beam column with six sides without reinforcement, we use $V_{CFRP} = 0$. The calculation formula of the shear bearing capacity of the core area of the other joints strengthened with the CFRP [18] is as follows:

$$V_{CFRP} = \sum_{i}^{n_0} \alpha_i A_{cfrp,i} \varepsilon_{cfrp,i} E_{cfrp,i} \tag{6}$$

where n_0 = number of CFRP categories. $A_{cfrp,i}$ = effective area of the joint core area strengthened with the CFRP. $E_{cfpr,i}$ = modulus of elasticity of CFRP. $\varepsilon_{cfrp,i}$ = the effective strain of the CFRP when the specimen is destroyed, which is not greater than the limit deformation of the CFRP; and the specific size is determined by the coordination of the cross-section deformation or the test. When the CFRP of the beam ends or the concrete is crushed and loses its bearing capacity, the CFRP strain, $\varepsilon_{cfrp,i}$, is equal to the effective strain of the CFRP in the core area of the joint. α_i = the correction factor for the influence of the bonding angle of the CFRP on the shear capacity; in this case, $\alpha_i = \cos\theta$. θ is the angle between the direction of the force of the CFRP and the horizontal axis.

Hence, the shear strength of the seismic-damaged composite steel-concrete frame joint strengthened with the CFRP can be computed as follows:

The first case is an aseismic-grade frame of nine degrees of fortification intensity:

$$V_u \le \frac{0.609 f_a^{'} t_w h_w}{\lambda - 0.2} + \frac{1}{\gamma_{RE}}\left(0.9\eta_j f_t b_j h_j + f_{yv}^{'} A_{svj}\frac{a - a_s^{'}}{s}\right) + \sum_{i}^{m} \alpha_i A_{cfrp,i} \varepsilon_{cfrp,i} E_{cfrp,i} \tag{7a}$$

Other cases:

$$V_u \le \frac{0.609 f_a^{'} t_w h_w}{\lambda - 0.2} + \frac{1}{\gamma_{RE}}\left(1.1\eta_j f_t b_j h_j + 0.05\eta_j N\frac{b_j}{b_c} + f_{yv}^{'} A_{svj}\frac{a - a_s^{'}}{s}\right) + \sum_{i}^{m} \alpha_i A_{cfrp,i} \varepsilon_{cfrp,i} E_{cfrp,i} \tag{7b}$$

4.3. Material damage description

The seismic pre-damage of the test specimen is described in term of strength reduction, which can be described by the strength reduction factor, a_F; the expression is as follows [24]:

$$a_F = 1 + \beta_1 D + \beta_2 D^2 \tag{8}$$

In Eq. (8), D is the damage index; β_1 and β_2 are the correlation coefficients, which are calculated using Eq. (9).

$$\beta_1 = 0.127 - 0.000586 f_{yv}^{'} + 0.229\left(\frac{f_{yv}^{'}}{1000}\right)^2 + 0.00143 f_c \tag{9a}$$

$$\beta_2 = -1.013 + 0.585n - 1.762n^2 + 0.183\rho_a + \frac{10.959}{f_c} \tag{9b}$$

where f_c is the axial compressive strength of concrete, n is the axial compression ratio, and f_a is the steel section ratio.

The strength degradation coefficient is used in the material description of the seismic-damaged SRC structures and the relationship between the strength degradation coefficient, a_F, and the structural material is obtained [1]:

$$f_{yv}^{'} = a_F f_{yv} \tag{10a}$$

$$f_{ck}^{'} = a_F f_{ck} \tag{10b}$$

$$f_a^{'} = a_F f_a \tag{10c}$$

where ρ_w is the stirrup ratio. f_{yv}, f_{ck}, and f_a are the strength of the components of the stirrup, the longitudinal reinforcement, and the steel section before an earthquake, respectively. $f_{ck}^{'}$ is the residual strength of the longitudinal reinforcement of the components after an earthquake.

4.4. Evaluation of the computational models

Table 4 shows the comparison of the theoretical value and measured values of the shear capacities of the specimens. The results show that the theoretical value agrees well with the measured value (not including the results of Zeng [25]). A formula for calculating the shear strength is proposed based on the beam's plastic hinge mechanism. The SRC beam structure is used in the Zeng specimen and the ultimate shear capacity of the SRC structure is larger than the ultimate bearing capacity of the RC structure. The results in Table 4 demonstrate that the influence of the steel ratio, seismic damage, reinforcement, and specific geometry on the shear strength of the structures should not be neglected. The calculated value of the shear capacity of the RC structure is close to the experimental value (the formula is put forward by GB 50010-2010 [19]). When the seismic damage is not considered, the reinforcement has a greater effect on the shear strength of the structure. The results are in agreement with the results of Yang et al. [29].

5. Summary and conclusions

Four composite steel-concrete frame joints were built to investigate the seismic behavior of seismic-damaged composite steel-concrete frame joints strengthened with CFRP sheets. The effects of the degree of seismic damage and the CFRP amount on the shear strength, ductility, energy dissipation, and stiffness degradation were critically researched. Based on the existing code, an expression to predict the shear strength of seismic-damaged composite steel-concrete frame joints strengthened with CFRP is presented; the expression considers the influences of the degree of seismic damage and amount of CFRP on the shear strength. The results of the proposed expression and the test results were compared. The following conclusions were drawn:

(1) The reinforcement of the joints with the CFRP sheets improved the ductility. The greater the degree of seismic damage, the better the reinforcement effect of the CFRP was. The CFRP-reinforced specimen reached or even exceeded the level of the original seismic state prior to the seismic damage up to a certain damage level.
(2) The stiffness degradation decreased less for the specimens strengthened with the CFRP than for the control specimen, indicating that the CFRP effectively prevented damage to the specimens and improved their ability to resist seismic deformation. Both the bearing capacity and deformation capacity of the severely damaged specimen were slightly better for the specimens strengthened with the CFRP.
(3) Compared to the original specimen, the average ultimate bearing capacity of the specimens strengthened with the CFRP increased by 15.58%, 13.27%, and 10.35%, respectively and the average ductility coefficient increased by 25.39%, 20.69%, and 5.33%, respectively. The results indicated that the CFRP sheets effectively improved the ductility, bearing capacity, and energy dissipation capacity of the specimens.

Table 4
Comparison of the theoretical value and measured value shear capacities of the specimens.

Specimen	Reference	Specimen section	CFRP layer	Seismic damage	*n*	Strength grade of concrete	Theoretical value/Experimental value
SRCC-0	Experimental data	SRC column-RC beam	–	–	0.4	C40	0.90
SRCC-1			2	–	0.4	C40	0.95
SRCC-2			2	Moderately damaged	0.4	C40	0.91
SRCC-3			2	Severely damaged	0.4	C40	0.93
J-1	Zeng [25]	SRC column-SRC beam	–	–	0.2	C60	0.61
J-2			–	–	0.4	C60	0.63
J-3			–	–	0.6	C60	0.57
J-4			–	–	0.6	C80	0.61
J-5			–	–	0.6	C100	0.59
SJ-1	Ma [26]	RC column-RC beam	–	–	0.25	C25	0.93
JSJ-2			1	–	0.25	C25	0.78
JSJ-3			1	–	0.25	C25	0.79
JSJ-4			2	–	0.25	C25	0.92
JSJ-5			2	–	0.25	C25	0.91
J-0	Lu et al. [27]	RC column-RC beam	–	–	0.5	C25	0.85
J-1			2	damaged	0.5	C25	0.89
J-2			2	damaged	0.5	C25	0.91
J-3			2	damaged	0.5	C25	0.87
J-4			2	–	0.5	C25	0.90
J10	Lee et al. [28]	RC column-RC beam	–	–	–	C30	0.94
J11			2	–	–	C30	0.91
J12			2	–	–	C30	0.92
SJ-1	Chang et al. [9]	RC column-RC beam	2	Damaged	0.15	C30	0.89
SJ-2			2	–	0.25	C30	0.83
SJ-3			2	Damaged	0.25	C30	0.85
SJ-4			2	Damaged	0.30	C30	0.90
SJ-5			2	Damaged	0.15	C30	0.87
SJ-6			2	Damaged	0.25	C30	0.81
SJ-7			2	Damaged	0.30	C30	0.88
SJ-8			2	–	0.25	C30	0.92
Mean							0.84

(4) Taking into account the influences of the degree of seismic damage and amount of CFRP on the shear strength, an expression to predict the shear strength of the seismic-damaged composite steel-concrete frame joints strengthened with CFRP was presented. The results were compared with the experimental results and were in agreement. The results of this study provide a reference for engineering practical applications.

Declaration of competing interest

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. All authors, which includes Chengxiang Xu, Sheng Peng, Chenfei Wang, and Zuotao Ma have approved the manuscript, including authorship and order of authorship, and agree with submission to Construction and Building Materials. This research is funded by National Natural Science Foundation of China (CN) (Grant No.51478048, 51678457), Natural Science Foundation of Hubei Province (innovation group) of China (CN) (Grant No. 2015CFA029) and their support is gratefully acknowledged. The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

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